RUPIN KHADWAL

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Education

University of Toronto – Engineering Science, Toronto ON **Expected Graduation 2024**

- Pursuing a Bachelor of Engineering with a specialization in Machine Intelligence
- Relevant coursework includes *Introduction to Computer Programming*, *Computer Algorithms* and Data Structures, *Python for Data Learning and Visualization*

Work Experience

Walmart Canada – Toronto ON

Digital & Efficiency AI Engineer **May 2022 – Present**

- Collaborating with various teams such as Marketing, Finance, Replenishment etc. to create an efficient promotions process via an AI-based approach
- Forging a diverse range of tools to compare different states of the Promotional Process of the company
- Mobilizing the use of AI-reliant tools in the workplace by creating a Python-based tool to completes price entries
- Working with Data Science to efficiently create a product recommendation tool to aid Merchants

Max-Planck Institute for the Structure and Dynamics of Matter - Hamburg, Germany

Research Associate / Software Developer **May 2021 – Present**

- Cultivating a GUI for the University of Hamburg and MPI to use for virtual experiments
- Creating a server-client-based application that handles client data and user input with different hardware devices
- Coding in Python to create numerous protocol buffers to send and receive data to and from a created server
- Implementing a database into a server-based system to handle several IP addresses, ports, and other client data

UTESCA, University of Toronto Engineering Consulting Association – Toronto ON

Team Lead **September 2020 – May 2021**

- Created an AI-based application that assesses the safety of buildings to reduce the number of bird collisions in collaboration with FLAP CANADA
- Collaborated and consulted with clients and industry professionals in a one-on-one environment
- Presented the final CNN and validated the effectiveness of the product via a test and validation dataset
- Successfully used programming languages such as Python, HTML, and CSS to cultivate the project

Projects

Braille AI Recognition Project

- Collected and scraped Braille characters in 4 different languages to create a validation, test, and training dataset
- Transformed and pre-processed all data using Pandas and Sklearn to fit the Neural Network's parameters
- Trained an image recognition CNN using Pytorch to recognize Braille characters in different languages

Movie Review Classification Project

- Utilized a Natural Language Processing and an IMDB review dataset to One Hot Encode reviews
- Constructed a Gated Recurrent Unit Cell to conduct a sentiment analysis
- Compared the Stochastic Gradient Descent to Adam optimization
- Built an RNN that outputs the general sentiment (positive or negative) of a review to a dependable accuracy

Recidivist Predictor Project

- Analyzed the COMPAS dataset to examine the probability of an offender to commit another offense
- Examined the False Positive Parity of predicting recidivism based on race
- Created an Adversarial Learning Procedure to correctly predict data

Relevant Modules: Pandas, PyTorch, Numpy, Random, Tensorflow, Matplotlib, Torchvision, Keras, JAX, OS, Time

Skills & Qualifications

- Outstanding ability to work with different audiences and stakeholders in professional environments
- Proficient in Python, C, Matlab, Java, Jupyter Labs, and Microsoft Office (Word, Excel, Outlook, Powerpoint)
- Fluent in English, Intermediate in French and Hindi

REFERENCES AVAILABLE UPON REQUEST